Signal iOS 8.15 (build 1655)

Source-to-binary verification of the App Store build

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Result. The compiled code (`__TEXT,__text`) of all 23 binaries in the App Store app is **byte-identical** to a local build of tag `8.15.0.1655` with Xcode 26.5 (17F42). That covers Signal's five own binaries and all 18 bundled frameworks. Every other difference falls into a category already explained in the 8.14 report: signature size, UUID, linker GOT-slot order (with every stub and pointer resolving to the same target), build metadata in plists, and bitmaps rasterised at build time from identical sources.

1 Method

This is the same method as 8.35.1 and 8.14, which describe it in full.

1. The source is cloned at the Xcode Cloud path `/Volumes/workspace/repository`.
2. `make dependencies` fetches the dependencies.
3. An unsigned `xcodebuild archive` runs with the scheme's App Store Release configuration, using the exact Xcode build recorded in the shipped `Info.plist`.
4. `verify/compare_build.sh` and `verify/explain_residuals.sh` compare the result with the decrypted App Store IPA.

Input	Value
App Store app	`org.whispersystems.signal_8.15.decrypted.ipa` from BarbossHack/reproducible-LFS, SHA-256 `7629023a…6ad7e4`. Version 8.15, build 1655. Every code-signature page outside the FairPlay-encrypted ranges matches Apple's signature.
Source	Tag `8.15.0.1655`, commit `a619caa`, the commit recorded in the shipped `Info.plist`. Pods submodule `1a4acf8`.
Toolchain	Xcode 26.5 (17F42), iOS SDK 26.5 (23F73), matching the shipped `DTXcodeBuild` and `DTSDKBuild`.

2 Results

Check	Result
`__TEXT,__text`, all 23 binaries	Byte-identical (65.6 MB, hashed with Apple's `segedit` and `shasum`)
Functions, Signal's 5 binaries	144,193 of 144,193 match after address normalisation
Strings, symbols, linked libraries	Identical
Header bytes	`__LINKEDIT` size (signature) in all 23 binaries; `LC_UUID` in 13 binaries
GOT slot order	`Signal` (1,668 stub bytes), `SignalUI` (1,159), `LibMobileCoin` (4) and `SignalServiceKit` (19 relative pointers in `__const`). Every stub and pointer resolves to the same target.
Plists	26 differ, only in `BuildDetails` time, `BuildMachineOSBuild`, `CFBundleVersion` and `DTAppStoreToolsBuild`

Check	Result
Icons and asset catalogs	The two loose icons differ by at most 8/255 per channel. In the catalogs, 8 + 8 rasterised renditions differ; each comes from a vector source that is identical in both builds, or is a packed atlas of such images. The NSE catalog is identical.

These residual differences are the ones Telegram's `ipadiff.py` does not normalise. Run on this build, it flags exactly the seven files BarbossHack's 8.15 run reported: `Signal`, `SignalUI`, `SignalServiceKit`, `LibMobileCoin`, `Info.plist` and the two icons.

3 Limitations

As for earlier versions:

- The public source itself is not reviewed.
- Prebuilt dependencies (libsignal, RingRTC, WebRTC, LibMobileCoin) are checked against their pinned versions, not rebuilt from their own source.
- The causes of the UUID and rasterisation differences are inferred, not proven.

4 Reproducing

```
git clone https://github.com/signalapp/Signal-iOS.git /Volumes/workspace/repository
cd /Volumes/workspace/repository && git checkout 8.15.0.1655 && make dependencies
# xcodebuild archive with Xcode 26.5 (17F42), as in the 8.35.1 report, then:
verify/compare_build.sh org.whispersystems.signal_8.15.decrypted.ipa <built Signal.app>
versions/8.15/out/ipa_vs_build
verify/explain_residuals.sh org.whispersystems.signal_8.15.decrypted.ipa <built
Signal.app> versions/8.15/out/residuals
```

The decrypted IPA is attached to the `signal-8.15` release. The app is also in `versions/8.15/signal_dec/`, and all outputs are in `versions/8.15/out/`.